

Homework (Jalapeno)

- Q1.** An archaeologist finds a coin from year 400 BC (represented as -400 on a timeline). What is the absolute value $|-400|$ representing its distance from year zero?
(A) -400
(B) 0
(C) 400
(D) 1
(E) -1
- Q2.** A Roman scroll is found 12 meters below ground level (-12). Which expression correctly shows the distance from the scroll to the surface (0)?
(A) $-|12|$
(B) $|-12|$
(C) $-(-12)$
(D) $|0 - 12|$
(E) $|12|$
- Q3.** Evaluate the expression representing the total distance from zero of two artifacts located at 15 and -15 : $|15| + |-15|$.
(A) 0
(B) 15
(C) 30
(D) -30
(E) 1
- Q4.** If the age of a Pharaoh's reign is represented by $|x| = 45$ on a historical timeline relative to a major war, which values could x be?
(A) Only 45
(B) Only -45
(C) 0 and 45
(D) 45 and -45
- (E) No solution
- Q5.** Which of the following ancient ruins is the furthest distance from the timeline center (0)?
(A) $|-250|$
(B) $|100|$
(C) $|-300|$
(D) $|50|$
(E) $|-150|$
- Q6.** Simplify the distance between a statue at -8 meters and a pillar at 0 meters using absolute value notation.
(A) $|-8| = 8$
(B) $-|8| = -8$
(C) $|8| = -8$
(D) $|-8| = -8$
(E) $-|-8| = 8$
- Q7.** Which statement about the absolute value of a year in the BC era (negative numbers) is always true?
(A) It is always negative.
(B) It is always zero.
(C) It is always positive.
(D) It depends on the century.
(E) It equals the year itself.
- Q8.** An ancient map marks two treasures. Treasure A is at $|-20|$ and Treasure B is at $|18|$. Which is closer to the origin (0)?
(A) Treasure A
(B) Treasure B
(C) They are equal distance
(D) Cannot be determined
(E) The origin itself

Open Ended Questions (FRQ)

- Q9.** An artifact from the Stone Age is found at -2500 on a timeline, and a Bronze Age tool is at -1200 . Calculate the sum of their absolute values: $|-2500| + |-1200|$.
- Q10.** Explain using the definition of distance why $|-50|$ and $|50|$ both equal 50 in the context of a ship sailing 50 leagues East vs 50 leagues West from a port (0).

Chef's Tips

- Emphasize that absolute value measures magnitude, not direction.
- Use the timeline as a visual aid for negative values in BC eras.
- Verify students understand that absolute value can never result in a negative number.